

Curriculum Vitae

JOSHUA D. BRAKENSIEK, PH.D.

Department of EECS
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[Google Scholar](#), [DBLP](#), [arXiv](#)

EDUCATION

Stanford University

- Doctor of Philosophy in Computer Science 2018 – 2024
 - Co-advisors: Aviad Rubinsten and Moses Charikar
 - Thesis: “Higher Order MDS Codes”
- Master’s of Science in Computer Science 2018 – 2021
 - Qualification Exam Topic: “Automated Design of Error-Correcting Codes”

Carnegie Mellon University

- Master’s of Science in Mathematical Sciences 2016 – 2018
 - Honor’s thesis: “Polymorphic Inquiries: Promise Constraint Satisfaction and Beyond”
 - Advisor: Venkatesan Guruswami
- Bachelor’s of Science in Mathematical Sciences 2014 – 2018
 - Minor in Science, Technology, and Society
 - University and Mellon College of Science Honors, GPA: 4.00/4.00

EMPLOYMENT

University of California, Berkeley

- Postdoctoral Scholar in EECS 2024 – present
 - NSF Mathematical Sciences Postdoctoral Research Fellowship 2025 – present
 - Supervisor: Venkatesan Guruswami

¹Updated September 23, 2026

JOURNAL PUBLICATIONS

22. **Improved Constructions and Lower Bounds for Maximally Recoverable Grid Codes** (joint with Manik Dhar and Sivakanth Gopi). IEEE Transactions on Information Theory, 2026. ([arXiv:2509.15013](#), [DOI](#))
21. **Separating MAX 2-AND, MAX DI-CUT, and MAX CUT** (joint with Neng Huang, Aaron Potechin, and Uri Zwick). SIAM Journal on Computing, 2026. **Special Issue.** ([arXiv:2212.11191](#), [GitHub](#), [DOI](#))
20. **Rigidity Matroids and Linear Algebraic Matroids with Applications to Matrix Completion and Tensor Codes** (joint with Manik Dhar, Jiyang Gao, Sivakanth Gopi, and Matt Larson). Combinatorica, 2026. ([arXiv:2405.00778](#), [DOI](#))
19. **Maximal Recoverability: A Nexus of Coding Theory** (joint with Venkatesan Gurusawmi). IEEE BITS, 2025. Special Issue on Error-Correcting Codes. ([arXiv:2602.22042](#), [DOI](#))
18. **Kapranov Degrees** (joint with Christopher Eur, Matt Larson, and Shiyue Li). International Mathematics Research Notices, 2025. ([arXiv:2308.12285](#), [DOI](#))
17. **SDPs and Robust Satisfiability of Promise CSP** (joint with Venkatesan Gurusawmi and Sai Sandeep). Discrete Analysis, 2025. ([arXiv:2211.08373](#), [DOI](#))
16. **Generalized GM-MDS: Polynomial Codes are Higher Order MDS** (joint with Manik Dhar and Sivakanth Gopi). IEEE Transactions on Information Theory, 2025. ([arXiv:2310.12888](#), [DOI](#))
15. **AG Codes Achieve List Decoding Capacity Over Constant-sized Fields** (joint with Manik Dhar, Sivakanth Gopi, and Zihan Zhang). IEEE Transactions on Information Theory, 2025. ([arXiv:2310.12898](#), [DOI](#))
14. **On the Mysteries of MAX NAE-SAT** (joint with Neng Huang, Aaron Potechin, and Uri Zwick). SIAM Journal on Discrete Mathematics, 2025. ([arXiv:2009.10677](#), [DOI](#))
13. **Generic Reed-Solomon Codes Achieve List-decoding Capacity** (joint with Sivakanth Gopi and Visu Makam). SIAM Journal on Computing, 2024. **Special Issue.** ([arXiv:2206.05256](#), [DOI](#))
12. **Improved Field Size Bounds for Higher Order MDS Codes** (joint with Manik Dhar and Sivakanth Gopi). IEEE Transactions on Information Theory, 2024. ([arXiv:2212.11262](#), [GitHub](#), [DOI](#))
11. **Robust Factorizations and Colorings of Tensor Graphs** (joint with Sami Davies). SIAM Journal on Discrete Mathematics, 2024. ([arXiv:2207.08913](#), [DOI](#))
10. **A Family of Dictatorship Tests with Perfect Completeness for 2-to-2 Label Cover** (joint with Venkatesan Gurusawmi). Chicago Journal of Theoretical Computer Science, 2023. ([ECCC TR17-141](#), [URL](#))
9. **Conditional Dichotomy of Boolean Ordered Promise CSPs** (joint with Venkatesan Gurusawmi and Sai Sandeep). TheoretiCS, 2023. ([arXiv:2102.11854](#), [DOI](#))

8. **Lower Bounds for Maximally Recoverable Tensor Codes and Higher Order MDS Codes** (joint with Sivakanth Gopi and Visu Makam). IEEE Transactions on Information Theory, 2022. ([arXiv:2107.10822](#), [DOI](#))
7. **The Resolution of Keller’s Conjecture** (joint with Marijn Heule, John Mackey, and David Narváez). Journal of Automated Reasoning, 2022. **Special Issue**. ([arXiv:1910.03740](#), [DOI](#))
6. **Constraint Satisfaction Problems with Global Modular Constraints: Algorithms and Hardness via Polynomial Representations** (joint with Sivakanth Gopi and Venkatesan Guruswami). SIAM Journal on Computing, 2022. ([arXiv:1902.04740](#), [DOI](#))
5. **The Quest for Strong Inapproximability Results with Perfect Completeness** (joint with Venkatesan Guruswami). ACM Transactions on Algorithms, 2021. ([ECCC TR17-080](#), [DOI](#))
4. **Promise Constraint Satisfaction: Algebraic Structure and a Symmetric Boolean Dichotomy** (joint with Venkatesan Guruswami). SIAM Journal on Computing, 2021. ([arXiv:1704.01937](#), [DOI](#))
3. **The Power of the Combined Basic LP and Affine Relaxation for Promise CSPs** (joint with Venkatesan Guruswami, Marcin Wrochna, and Stanislav Živný). SIAM Journal on Computing, 2020. ([arXiv:1907.04383](#), [DOI](#))
2. **Efficient Low-Redundancy Codes for Correcting Multiple Deletions** (joint with Venkatesan Guruswami and Samuel Zbarsky). IEEE Transactions on Information Theory, 2017. ([arXiv:1507.06175](#), [DOI](#))
1. **Efficient Geometric Probabilities of Multi-Transiting Exoplanetary Systems from CORBITS** (joint with Darin Ragozzine). The Astrophysical Journal, 2016. ([arXiv:1602.07014](#), [GitHub](#), [DOI](#))

REFEREED CONFERENCE PUBLICATIONS

33. **The Richness of CSP Non-redundancy** (joint with Venkatesan Guruswami, Bart M.P. Jansen, Victor Lagerkvist, and Magnus Wahlström). FOCS 2026 (to appear). ([arXiv:2507.07942](#))
32. **Super-linear Lower Bounds for CSP Non-Redundancy via Shrinking Instances** (joint with Venkatesan Guruswami, Bart M.P. Jansen, Victor Lagerkvist, and Magnus Wahlström). IPEC 2026. **Best Paper**. ([arXiv:2605.19055](#))
31. **Classification of Non-redundancy of Boolean Predicates of Arity 4** (joint with Venkatesan Guruswami and Aaron Putterman). CP 2026. ([arXiv:2603.21353](#), [DOI](#))
30. **Multiplicative Error Set System Sparsification: A Simpler Proof via Chain Length Contraction** (joint with Venkatesan Guruswami and Aaron Putterman). ICALP 2026. ([arXiv:2605.01508](#), [DOI](#))
29. **Unique Decoding of Reed–Solomon and Related Codes for Semi-Adversarial Errors** (joint with Yeyuan Chen, Manik Dhar, and Zihan Zhang). ICALP 2026. ([arXiv:2504.10399](#), [DOI](#))

28. **Improved Approximation Algorithms for Multiway Cut by Large Mixtures of New and Old Rounding Schemes** (joint with Neng Huang, Aaron Potechin, and Uri Zwick). STOC 2026. ([arXiv:2603.28700](#), [DOI](#))
27. **Combinatorial Bounds for List Recovery via Discrete Brascamp–Lieb Inequalities** (joint with Yeyuan Chen, Manik Dhar, and Zihan Zhang). STOC 2026. ([arXiv:2510.13775](#), [DOI](#))
26. **From Random to Explicit via Subspace Designs with Applications to Local Properties and Matroids** (joint with Yeyuan Chen, Manik Dhar, and Zihan Zhang). STOC 2026. ([arXiv:2510.13777](#), [DOI](#))
25. **New Algorithms and Hardness Results for Robust Satisfiability of (Promise) CSPs** (joint with Lorenzo Ciardo, Venkatesan Guruswami, Aaron Potechin, and Stanislav Živný). SODA 2026. ([arXiv:2602.10368](#), [DOI](#))
24. **MAX BISECTION Might Be Harder to Approximate than MAX CUT** (joint with Neng Huang, Aaron Potechin, and Uri Zwick). SODA 2026. ([arXiv:2512.04951](#), [DOI](#))
23. **Redundancy Is All You Need** (joint with Venkatesan Gurusawmi). STOC 2025. ([arXiv:2411.03451](#), [DOI](#))
22. **AG Codes Achieve List Decoding Capacity Over Constant-sized Fields** (joint with Manik Dhar, Sivakanth Gopi, and Zihan Zhang). STOC 2024. ([arXiv:2310.12898](#), [DOI](#))
21. **Generalized GM-MDS: Polynomial Codes are Higher Order MDS** (joint with Manik Dhar and Sivakanth Gopi). STOC 2024. ([arXiv:2310.12888](#), [DOI](#))
20. **Tight Approximability of MAX 2-SAT and Relatives, under UGC** (joint with Neng Huang and Uri Zwick). SODA 2024. ([arXiv:2310.12911](#), [GitHub](#), [DOI](#))
19. **Separating MAX 2-AND, MAX DI-CUT and MAX CUT** (joint with Neng Huang, Aaron Potechin, and Uri Zwick). FOCS 2023. ([arXiv:2212.11191](#), [GitHub](#), [DOI](#))
18. **Improved Field Size Bounds for Higher Order MDS Codes** (joint with Manik Dhar and Sivakanth Gopi). ISIT 2023. ([arXiv:2212.11262](#), [GitHub](#), [DOI](#))
17. **SDPs and Robust Satisfiability of Promise CSP** (joint with Venkatesan Gurusawmi and Sai Sandeep). STOC 2023. ([arXiv:2211.08373](#), [DOI](#))
16. **Generic Reed-Solomon Codes Achieve List-decoding Capacity** (joint with Sivakanth Gopi and Visu Makam). STOC 2023. ([arXiv:2206.05256](#), [DOI](#))
15. **Conditional Dichotomy of Boolean Ordered Promise CSPs** (joint with Venkatesan Guruswami and Sai Sandeep). ICALP 2021. ([arXiv:2102.11854](#), [DOI](#))
14. **On the Mysteries of MAX NAE-SAT** (joint with Neng Huang, Aaron Potechin, and Uri Zwick). SODA 2021. ([arXiv:2009.10677](#), [DOI](#))
13. **Smoothed Complexity of 2-player Nash Equilibria** (joint with Shant Boodaghians, Sam Hopkins, and Aviad Rubinstein). FOCS 2020. ([arXiv:2007.10857](#), [DOI](#))
12. **Coded Trace Reconstruction in a Constant Number of Traces** (joint with Ray Li and Bruce Spang). FOCS 2020. ([arXiv:1908.03996](#), [DOI](#))

11. **The Resolution of Keller’s Conjecture** (joint with Marijn Heule, John Mackey, and David Narváez). IJCAR 2020. **Best Paper Award**. ([arXiv:1910.03740](#), [DOI](#))
10. **CSPs with Global Modular Constraints: Algorithms and Hardness via Polynomial Representations** (joint with Sivakanth Gopi and Venkatesan Guruswami). STOC 2020. ([arXiv:1902.04740](#), [DOI](#))
9. **Constant-factor Approximation of Near-linear Edit Distance in Near-linear Time** (joint with Aviad Rubinfeld). STOC 2020. ([arXiv:1904.05390](#), [DOI](#))
8. **Symmetric Polymorphisms and Efficient Decidability of Promise CSPs** (joint with Venkatesan Guruswami). SODA 2020. ([arXiv:1907.04383v1](#), [DOI](#))
7. **Bridging between 0/1 and Linear Programming via Random Walks** (joint with Venkatesan Guruswami). STOC 2019. ([arXiv:1904.04860](#), [DOI](#))
6. **An Algorithmic Blend of LPs and Ring Equations for Promise CSPs** (joint with Venkatesan Guruswami). SODA 2019. ([ECCC TR18-059](#), [DOI](#))
5. **Promise Constraint Satisfaction: Algebraic Structure and a Symmetric Boolean Dichotomy** (joint with Venkatesan Gurusawmi). SODA 2018. ([arXiv:1704.01937](#), [DOI](#))
4. **The Quest for Strong Inapproximability Results with Perfect Completeness** (joint with Venkatesan Gurusawmi). APPROX 2017. ([ECCC TR17-080](#), [DOI](#))
3. **Vertex Isoperimetry and Independent Set Stability for Tensor Powers of Cliques**. RANDOM 2017. ([arXiv:1702.04432](#), [DOI](#))
2. **New Hardness Results for Graph and Hypergraph Colorings** (joint with Venkatesan Guruswami). CCC 2016. ([ECCC TR16-029](#), [DOI](#))
1. **Efficient Low-Redundancy Codes for Correcting Multiple Deletions** (joint with Venkatesan Guruswami and Samuel Zbarsky). SODA 2016. ([arXiv:1507.06175](#), [DOI](#))

PREPRINTS

9. **Optimal Sparsifiers for Minkowski Sums and Sums of Seminorms** (joint with Arpon Basu, Yeyuan Chen, Aaron Putterman, Victor Reis, Zihan Zhang). Manuscript. ([arXiv:2609.20238](#))
8. **Separating Non-redundancy and Chain Length** (joint with Venkatesan Guruswami and Aaron Putterman). Manuscript. ([arXiv:2609.17914](#))
7. **Algorithmic List Decoding of Reed-Solomon Codes up to Capacity** (joint with Yeyuan Chen, Aaron Putterman, Zihan Zhang, and Kai Zhe Zheng). Manuscript. ([arXiv:2609.08005](#))
6. **Bounds and Limitations on Codes Achieving List Recovery Capacity** (joint with Yeyuan Chen, Aaron Putterman, and Zihan Zhang). Manuscript. ([arXiv:2607.19576](#))
5. **Quantum Cut Sparsifiers** (joint with Arpon Basu, Pravesh Kothari, and Aaron Putterman). Manuscript. ([arXiv:2606.09728](#))

4. **Many Hamiltonians Are Sparsifiable** (joint with Arpon Basu and Aaron Putterman). Manuscript. ([arXiv:2605.02211](#))
3. **Tight Bounds for Sparsifying Random CSPs** (joint with Venkatesan Guruswami and Aaron Putterman). Manuscript. ([arXiv:2508.13345](#))
2. **A Simple Sublinear Algorithm for Gap Edit Distance** (joint with Moses Charikar and Aviad Rubinfeld). Manuscript. ([arXiv:2007.14368](#))
1. **Bounds on the Size of Sound Monotone Switching Networks Accepting Permutation Sets of Directed Trees** (joint with Aaron Potechin). Manuscript. ([arXiv:1301.3780](#))

OTHER WRITINGS

2. **Automated Design of Error-Correcting Codes Part 2**. Theory Dish — Stanford's CS Theory Research Blog. ([Wordpress](#))
1. **Automated Design of Error-Correcting Codes Part 1**. Theory Dish — Stanford's CS Theory Research Blog. ([Wordpress](#))

SELECTED TALKS

18. *The Richness of CSP Non-redundancy*. FOCS 2026, New York, NY, November 2026 (upcoming).
17. *TBD*. FOCS 2026 workshop on Subspace Designs: A Universal Tool for Derandomization, New York, NY, November 2026 (upcoming).
16. *Algorithmic List Decoding of Reed-Solomon Codes up to Capacity*. Stanford Theory Lunch, Stanford, CA, October 2026 (upcoming).
15. *From Random to Explicit via Subspace Designs with Applications to Local Properties and Matroids*. Simons Institute for Theory of Computing, Berkeley, CA, September 2026 (upcoming).
14. *Improved Approximation Algorithms for Multiway Cut by Large Mixtures of New and Old Rounding Schemes*. STOC 2026, Salt Lake City, UT, June 2026.
13. *Redundancy is All You Need (for CSP Sparsification)*. Workshop on Algorithms for Large Data (WALDO), Online, April 2025.
12. *Improved Field Size Bounds for Higher Order MDS Codes*. ISIT 2023, Online, June 2023.
11. *Robust Algorithms for Promise CSPs*. Dagstuhl Seminar 22201, Online, May 2022.
10. *Some Interesting Minions*. Dagstuhl Seminar 22201, Online, May 2022.
9. *Constant-factor Approximation of Near-linear Edit Distance in Near-linear Time*. Stanford Theory Lunch, Online, December 2020.
8. *Smoothed Complexity of 2-player Nash Equilibria*. FOCS 2020, Online, October 2020.
7. *Smoothed Complexity of 2-player Nash Equilibria*. Microsoft Research MLO Seminar, Online, September 2020.

6. *Constant-factor Approximation of Near-linear Edit Distance in Near-linear Time*. STOC 2020, Online, June 2020.
5. *Symmetric Polymorphisms and Efficient Decidability of Promise CSPs*. SODA 2020, Salt Lake City, UT, January 2020.
4. *Coded Trace Reconstruction in a Constant Number of Traces*. CMU ACO Seminar, Pittsburgh, PA, September 2019.
3. *Coded Trace Reconstruction in a Constant Number of Traces*. Microsoft Research, Redmond, WA, July 2019.
2. *Bridging between 0/1 and Linear Programming via Random Walks*. STOC 2019, Phoenix, AZ, June 2019.
1. *An Algorithmic Blend of LPs and Ring Equations for Promise CSPs*. SODA 2019, San Diego, CA, January 2019.

PROFESSIONAL ACTIVITIES

Program Committee

- I served on the International Symposium on Mathematical Foundations of Computer Science (MFCS) 2026 program committee.

Referee Work

- I write 20+ referee reports a year for leading venues in computer science, mathematics, and engineering. Selected venues I have reviewed for (often multiple times) include
 - Journals: *Journal of the ACM*, *SIAM Journal on Computing*, *TheoretCS*, *European Journal of Combinatorics*, *Electronic Journal of Combinatorics*, *SIAM Journal on Matrix Analysis and Applications*, *Journal of Automated Reasoning*, *IEEE Transactions on Information Theory*, *Theory of Computing*, *ACM Transactions on Algorithms*, *ACM Transactions on Computation Theory*, *IEEE Communication Letters*, *SIAM Journal on Discrete Mathematics*
 - Conference Proceedings: *Symp. on Theory of Computing (STOC)*, *Symposium on Foundations of Computer Science (FOCS)*, *Symp. on Discrete Algorithms (SODA)*, *Computational Complexity Conference (CCC)*, *Intl. Colloquium on Automata, Languages, and Programming (ICALP)*, *Logic in Computer Science (LICS)*, *International Symposium on Information Theory (ISIT)*
 - Mathematical Reviews

Other

- I served as the webmaster of <https://theory.stanford.edu> from 2019 to 2023.
- I served as the undergraduate representative on the CMU MCS College Council in the 2017-18 academic year.

TEACHING EXPERIENCE

Stanford University

- Course Assistant for Incentives in Computer Science (CS 269I) taught by Aviad Rubinstein in Spring 2021. **Top 5% of CS department CAs.**
- Course Assistant for Unfulfilled Algorithmic Fantasies (CS 354) taught by Aviad Rubinstein in Spring 2019. **Top 5% of CS department CAs.**

Carnegie Mellon University

- Teaching Assistant for Discrete Differential Geometry (CS 15-458/15-858) taught by Keenan Crane in Fall 2017.

Mathematical Association of America

- Grader to help train and select the members of the USA IMO team 2014 – 2017

INDUSTRY EXPERIENCE

Microsoft Research, Redmond

- Intern in the algorithms group, supervised by Sivakanth Gopi. Summer 2020

Exprii, Inc.

- Web design technical consultant. 2014

SELECTED AWARDS

Research

- IPEC 2026 Best Paper Award 2026
- NSF Mathematical Sciences Postdoctoral Fellowship 2025
- FOCS 2023 SICOMP Special Issue Invitation 2023
- STOC 2023 SICOMP Special Issue Invitation 2023
- Microsoft Research PhD Fellowship 2021
- IJCAR 2020 Best Paper Award co-winner 2020
- IJCAR 2020 JAR Special Issue Invitation 2020
- NSF Graduate Research Fellowship 2018
- CRA Outstanding Undergraduate Researcher Award 2018
- Goldwater Scholarship 2016
- Davidson Fellow 2013
- Top Five Awardee: Written Research, Research Science Institute 2012

Academic Competitions

- Putnam Fellow (top 5 individual) and 1st place team 2016
- Putnam Competition 8th place individual and 2nd place team 2015
- Putnam Competition 10.5th place individual 2014
- Two-time gold medalist: International Olympiad in Informatics 2013 – 2014
- Silver medalist: International Mathematical Olympiad 2014
- USA Mathematical Olympiad sole perfect scorer (Samuel L. Greitzer/Murray S. Klamkin Award for Mathematical Excellence) 2014
- Akamai Foundation Scholarship 2014
- Two-time USA Mathematical Olympiad Winner (top 12) 2012, 2014

Other

- Carnegie Mellon University Senior Leadership Recognition 2018
- Phi Kappa Phi inductee 2018
- Phi Beta Kappa *early* inductee 2017
- Knaster-McWilliams Scholar 2014–2018
- Pi Mu Epsilon inductee 2012
- Photograph “Symphony of Architectural Geometry” of Chandler-Gilbert Community College was selected as the 2013, Week 52 photo in the MAA Found Math column of the Mathematical Association of America. The photograph was also featured in the MAA’s 100th anniversary calendar.